

Component Repair Plan

GBPUSD · measured on 2026-06-10 to 07-23, data never used for tuning · standalone directional accuracy at a 1-hour horizon

What the measurement found

Every component was scored on one question: **when it says BUY, does price actually rise?** Independent of the engine's direction choice and of any trade plan. 50% is a coin flip.

Pooled across all conditions, nothing exceeds 54.6% and nothing falls below 43.1% — the whole stack sits in a narrow band around chance. That pooled view is misleading, and it is why every attempt to fix the system by reweighting failed. **Split by market regime, the components separate into two groups that are accurate in opposite conditions.**

The two groups

Group	Accurate in	Wrong in	Components
Mean-reversion / value	TRENDING 56–70%	RANGING 43–47%	volume_profile, premium_discount, macd, fib_confluence, rsi_divergence, order_flow
Trend / structure	RANGING 52–57%	TRENDING 27–45%	market_structure, support_resistance, wyckoff, mtf_cascade, ob_mitigation, h1_direction

This is a clean inversion, not noise. The ranging cells carry 400–766 observations each and the trending-calm cells 250–464. Two independent groups of indicators, each reliably right where the other is reliably wrong, is a structural property — and it is directly actionable.

Per-component accuracy by regime

Green $\geq 55\%$, red $\leq 45\%$. Cells with fewer than 40 observations are omitted rather than shown as thin numbers.

Component	All regimes	Trend calm	Trend vol	Ranging	Squeeze
mean_reversion.volume_profile	54.6% n=723	60.2% n=269	70.0% n=70	43.6% n=259	60.9% n=87
trend.macd	52.2% n=880	56.4% n=257	61.2% n=49	51.8% n=415	42.6% n=136
smc.premium_discount	52.0% n=1610	57.3% n=464	62.6% n=99	46.5% n=764	57.5% n=233
smc.supply_demand	52.0% n=415	47.6% n=105	—	52.6% n=211	54.9% n=71
smc.fvg_ifvg	50.8% n=1612	48.5% n=464	47.5% n=99	51.8% n=766	51.9% n=233
smc.ob_mitigation	49.2% n=1612	46.6% n=464	40.4% n=99	52.5% n=766	44.6% n=233
trend.support_resistance	48.9% n=1610	44.3% n=463	38.4% n=99	54.1% n=765	43.3% n=233
trend.adx_ema_trend	48.9% n=1496	43.9% n=460	44.4% n=99	52.3% n=685	49.0% n=202
mean_reversion.round_numbers	48.4% n=281	40.2% n=82	—	53.7% n=121	50.9% n=57
smc.order_flow	48.3% n=1612	52.2% n=464	55.6% n=99	44.9% n=766	49.8% n=233
momentum.rsi_adaptive	48.0% n=408	53.1% n=113	—	40.1% n=217	—
pattern.elliott_wave	47.5% n=1248	44.8% n=319	50.7% n=67	49.3% n=619	46.0% n=200
momentum.rsi_divergence_informed	47.5% n=950	49.3% n=335	63.8% n=69	43.1% n=397	49.1% n=108
mean_reversion.fib_confluence	47.3% n=780	57.6% n=203	—	42.6% n=373	46.0% n=174
trend.mtf_cascade	47.0% n=1518	39.3% n=412	35.8% n=95	52.4% n=735	44.7% n=226
smc.market_structure	46.8% n=960	43.0% n=305	27.3% n=77	55.7% n=449	37.2% n=113
momentum.stochastic_raw	46.7% n=398	44.0% n=109	—	44.1% n=186	60.7% n=61
pattern.candlestick	46.5% n=518	44.3% n=158	—	45.7% n=234	47.4% n=78
trend.h1_direction	45.8% n=743	46.2% n=234	32.4% n=71	49.5% n=323	37.4% n=91
pattern.wyckoff	45.0% n=522	45.2% n=299	33.7% n=89	57.1% n=98	—
momentum.stochastic_adaptive	43.1% n=332	46.3% n=95	—	35.1% n=171	—

The repair plan

1 — Gate every component on regime

This is the single change with the most measured support. A component should only vote where it is measurably accurate, and should be silenced where it is measurably wrong. Under the design spec's own rules a silenced member contributes zero at its assigned weight — silence dilutes, it is never redistributed — so this needs no new mechanism, only a regime condition on each member's participation.

In trending conditions the six mean-reversion components vote and the six trend/structure components are silent. In ranging conditions the assignment reverses. On the measured cells this replaces a ~48% blended read with one built only from components running 55–70% in the regime where they are used.

2 — Remove the duplicate trend votes

wyckoff, **adx_ema_trend**, **mtf_cascade** and **h1_direction** all answer the same question — which way has price been moving. Reading the source confirms it: `_detect_wyckoff_phase` returns MARKUP when trend is BULLISH and MARKDOWN when BEARISH. That is a trend label carrying a Wyckoff name, not an accumulation/distribution analysis. Four votes for one piece of evidence violates the architecture's own no-duplicate-representation rule, and all four measure below chance overall (45.0%, 48.9%, 47.0%, 45.8%).

Keep one trend representative, gated on regime per item 1. Retire the other three or demote them to modifiers.

3 — Rebuild around volume_profile

It is the strongest component measured: 54.6% pooled, and 60.2% / 70.0% / 60.9% in trending-calm, trending-volatile and squeeze. It was also the most consistent single indicator across every earlier test in this project. It deserves to be the anchor of the directional read rather than one vote among twenty, most of which are below chance.

4 — Fix the components that are wrong, not just weak

market_structure at 27.3% and **h1_direction** at 32.4% in trending-volatile are not weak signals — they are close to reliably inverted, which normally indicates a defect rather than an absent edge. These warrant the same source audit that exposed the Wyckoff duplication. Note that simple sign inversion was tested and failed, so the repair must come from understanding the mechanism, not from flipping the output.

What this plan will and will not deliver

Claim	Status
Components are strongly regime-dependent	MEASURED — large samples, unseen data
Regime gating should beat the pooled read	EXPECTED — not yet tested out-of-sample
Reaches roughly 55-60% directional accuracy	PLAUSIBLE — implied by the measured cells
Reaches 65% win rate at 1:2 reward/risk	NOT SUPPORTED by any measurement here

Directional accuracy is not win rate. A component that calls direction correctly 60% of the time still has to survive a stop before its target, and on this instrument only 37.5% of setups ever travel 2R. Regime gating raises the quality of the direction call; it does not move that ceiling.

Required validation protocol

The regime split above was **found** on this segment, so testing it on the same segment would be circular and would produce a confident, false result. During this project roughly forty configurations were tested against a single sample; eight looked strong on one window and died on the next, and three pre-registered hypotheses that passed on searched data failed on unseen data.

So this plan is not validated by re-running it here. It requires: implement regime gating, pull history that no analysis has touched, and test once with decision rules written before the data is opened. Anything else repeats the error this document exists to avoid.

Measurements: 5,899 replayed decisions, GBPUSD M15, 2026-06-10 to 07-23 (segment never used for tuning). Accuracy is the fraction of directional calls followed by favourable price movement over 60 M1 bars. Regime labels are the engine's own `trading_regime.state` at decision time.